The Core Idea Behind Transformers & LLMs

Deep Learning	NLP	LLMs	Transformers	RAG

What Is Attention?

Attention is a mechanism that lets a neural network focus on the most relevant parts of an input when producing each output token. Instead of compressing an entire sequence into one fixed vector, attention dynamically assigns a weight to every input position — giving more influence to positions that matter for the current step.

Where Does It Fit?

- **Raw AI / Deep Learning** — Attention is a fundamental DL building block, not specific to RAG or LangChain.
- **Transformers (GPT, BERT, etc.)** — Every modern LLM is built entirely on attention layers ("Attention Is All You Need", Vaswani et al. 2017).
- **RAG** — The LLM inside a RAG pipeline uses attention internally to fuse retrieved chunks with the query.
- **LangChain** — LangChain orchestrates LLMs; attention is what makes those LLMs understand context.

How Scaled Dot-Product Attention Works

Each token is projected into three vectors: Query (Q), Key (K), and Value (V). The attention score between a query and all keys is computed, scaled, softmaxed, then used to weight the values:

Formula:

$$\text{Attention}(Q, K, V) = \text{softmax}(Q \cdot K^T / \sqrt{d_k}) \cdot V$$

d_k	Dimension of the key vectors. Dividing by √d_k prevents the dot products from growing too large and causing vanishing gradients in softmax.

Multi-Head Attention

Instead of one attention function, the model runs h parallel attention heads each learning different relationships (syntax, coreference, semantics). Outputs are concatenated and projected:

```
MultiHead(Q,K,V) = Concat(head_1, ..., head_h) · W_O
```

Types of Attention

Type	Description	Used In
Self-Attention	Each token attends to all other tokens in the same sequence	Transformer encoder/decoder
Cross-Attention	Query from one sequence, K/V from another	Decoder attending to encoder
Causal/Masked	Future tokens masked so model can't peek	GPT-style autoregressive LLMs
Sparse Attention	Only attend to a subset to save memory	Long documents (Longformer)

Why It Matters for You (LLMs & RAG)

- Understanding attention helps you write better prompts — place the most important context near the query.
- In RAG, retrieved chunks compete for the model's attention; chunk quality and ordering directly affect output.
- Context window limits are attention limits — the model can only "attend" to tokens within its window.
- Fine-tuning, LoRA, and PEFT all modify how attention weights are updated.